

Classroom physics

June 2025 | Issue 73

The magazine for IOP affiliated schools and colleges

Understanding the weather

The atmospheric physics issue

Our electric atmosphere

Clouds and climate change

Weather-themed experiments for the classroom

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Our cover for this issue is taken from the photograph 'Saturated Earth and Sky's Promise' by Ellis Skelton. This image, shown here in full, won Second Place in the Standard Chartered Weather Photographer of the Year 2024 competition. Our thanks to photographer Ellis Skelton, and Nathan Reece from RMetS.

Credit: © Ellis Skelton

Editors' note

This issue, on the theme of atmospheric physics, is made in partnership with the Royal Meteorological Society (RMetS).

With summer term finally here and the end of the year in sight, hopefully we're all enjoying some sunshine. Weather is a huge part of life – for people in the UK and Ireland, almost to the point of obsession – and of increasing concern as weather patterns adjust to the changing climate. It's also full of fascinating physics.

We've put together a set of weather-related resources in our pull-out section (pages 9–12), looking at clouds and wind. As we note on page 9, there's a large degree of crossover with the geography curriculum, but physicists and geographers can have different ways of understanding and explaining some of the processes driving weather phenomena.

The theme continues in our feature stories, which are both written by experts on the weather. On page 6, Dr Keri Nicoll explains how clouds power a giant electric circuit which flows through the atmosphere. On page 7, Professor Ellie Highwood looks at clouds in more detail, unpicking some of the challenges they pose to physicists looking to model the atmosphere.

And in our digest section, we've got some great contributions from a range of organisations and individuals with useful resources and tips to share for teachers and technicians thinking about this topic. Thanks, as ever, to everyone who has contributed.

Have a great end to the year.

Sylvia Knight (Head of Education at RMetS), Dan Watson and Taj Bhutta

education@iop.org

Editors

Sylvia Knight
Dan Watson
Taj Bhutta

Design

Sophie Kate Newton

education@iop.org

IOP affiliated schools and colleges will receive with this issue...

A booklet from the Royal Meteorological Society highlighting careers in weather and climate.

Also available to download at metlink.org/careers



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IOP activities round-up

The IOP Learning & Skills and Policy & Public Affairs teams were busy during the spring term with a number of responses to government activities, as part of IOP's ongoing work to champion issues affecting physics education and physics teachers.

The IOP's response to the Government's interim curriculum review, while welcoming its recognition of systemic problems that lead to uneven outcomes for children across different schools and regions, was critical of how the review has so far failed to consider how these challenges affect specific subjects.

In a statement, IOP Director of Science, Innovation and Skills Louis Barson said: "[T]he current approach to structuring multiple different pathways through the sciences at GCSE compounds the disadvantage faced by certain groups of young people already facing socioeconomic disadvantage. As a result, too many young people are put off, or denied, the opportunity to study physics and physics-related pathways post-16, harming their life chances and holding back our economy."

(Read the full response here: bit.ly/InterimCurriculumReview)

The IOP also made submissions to other government consultations and parliamentary inquiries. The Learning & Skills and Policy & Public Affairs teams provided evidence to the Public Accounts Committee's inquiry about increasing teacher numbers for secondary schools and FE colleges; to Ofsted's consultation on the way it inspects education; and to the Department for Education's consultation on school accountability reform.

ITE tutors and honorary fellows

The IOP hosted a meeting of initial teacher education (ITE) tutors on 30 April at its London offices.



Credit: Shutterstock

By working closely with the ITE community, we can champion teacher training with government, and flag the issues and challenges it faces, working towards the best possible experience for all physics trainees.

Among four IOP honorary fellows announced in March, Professor Averil MacDonald was bestowed the honour for her lifelong commitment to equality, diversity and inclusion, and her work promoting increased girls' participation in STEM. She has also advocated for green technologies, and wrote an article for the September 2024 issue of Classroom Physics about hydrogen fuel cells.

Ongoing curriculum improvement work

In Scotland, IOP officers have continued to be closely involved in the Scottish Government's curriculum development work. The Core Group of advisers for the Curriculum Improvement Cycle for the sciences includes IOP learning and skills manager, Stuart Farmer, and IOP coach, Martyn Crawshaw. The group met for a four-day meeting in March to begin the process of identifying the 'Big Ideas' for the sciences curriculum in Scotland. This drew upon IOP's 'Fundamentals of 11 to 19 physics'. The larger Collaboration Group meets in June to consider the Core Group's outputs. The Core Group is scheduled to meet for a further eight days through 2025–26 to take this work forward.

The IOP Scotland team also recently completed the 50th annual Stirling meeting for physics teachers in Scotland. There were over 100

attendees, and we'll share a fuller report from the landmark event in the next issue.

In Northern Ireland, the IOP team have engaged with the ongoing review of education, responding to an open consultation and meeting with Lucy Crehan, the educational consultant heading up the initial stage of the review. Work in this area draws on the recommendations of the Primary Curriculum Advisory Group, as well as the 'Fundamentals of 11 to 19 physics' framework.

The annual Frontiers of Physics conference for teachers of physics in Ireland will be held on Saturday 20 September in the IOP building at 35 Westland Row, Dublin. A save the date and details of how to purchase tickets will be available shortly.

Boosting science education in Wales

A collaborative programme by IOP Wales, the Royal Society of Chemistry and Science Made Simple has secured funding from the Welsh Government's Curriculum for Wales grant to enhance science teaching in primary and secondary schools in Wales. The programme will strengthen practitioners' pedagogical and subject knowledge and build confidence through evidence-based professional development and specialised mentoring and coaching. Practitioners and technicians will benefit from tailored teaching resources and online modules, alongside the creation of networks to foster peer support. More information can be found here: iop.org/boosting-science-education-wales



IOP-UCL research assesses impact of spaced learning in physics

A new paper on spaced learning in physics, co-written by IOP teacher educators and neuroscientists from UCL, has found that the technique can double the impact of physics teaching.

Spaced learning is a form of teaching that intersperses intense learning with timed breaks in which students undertake unrelated activities. Neuroscientists have found the distraction periods to be beneficial because they help the brain's synapses to reset quickly. This enables the brain to form longer-term memories of the content on repeated viewing.

While spaced learning has been becoming more common in UK schools, there has been a shortage of research into its effectiveness in physics.

The researchers, led by IOP's Alessio Bernardelli, created a 50-minute video on atomic structure that condensed content from 8–12 hours' worth of conventional teaching. The film contains three learning sections, with two distraction sections in

between. Six non-selective state schools took part in a trial, in which 336 students watched the film and completed tests, which were marked by an independent physics examiner.

The results were found to double the impact of traditional teaching for students who watched the video and then received classroom teaching. On its own, the video enabled the same level of progress two months later as traditional teaching of GCSE physics.

Rachel Hartley, a physics teacher coach for the IOP and one of the report's authors, wrote an article for the IOP website explaining the importance of this work. She wrote: "These findings could be transformative for physics education, especially for schools without specialist physics teachers."

The technique could speed up subject knowledge teaching and free up time in the classroom for activities to give a deeper understanding of concepts, such as practical work and tackling lingering misconceptions.

Find out more and read Rachel's article at bit.ly/SpacedLearningBlog

School subsidy increased for physics subject knowledge programme

The Subject Knowledge for Physics Teaching (SKPT) CPD programme, run by the Ogden Trust since 2022, has had its funding from the Department for Education extended. The decision makes more funding available for school subsidies. On successful completion of a module, schools will now receive up to £600 for the cost of teacher cover.

"We know that releasing teachers for CPD and the cost of teacher cover can be a barrier to participation," says Charley Phillips, Ogden Trust Head of Teacher Support. "But we also know that the proportion of physics teaching hours taught by physics specialists has fallen by three percentage points over the last six years and that physics teacher recruitment continues to fall short of the government targets. We hope the increase in subsidy will enable more teachers to take part in SKPT, developing their subject knowledge and classroom confidence."

Accredited by the IOP, the modules require up to 20 hours of learning each, including face-to-face sessions (covering the required practicals), webinars and independent online work. Since 2022, more than 1,350 modules have been completed.

The new DfE funding of £2.8 million is supporting the delivery of 1,800 modules between 1 April 2025 and 31 March 2026.

The modules are available free of charge to qualified teachers who teach physics in an English state school but did not complete their QTS in a physics specialism.

For more information and booking, visit stem.org.uk/SKPT



New UK investment in girls' maths

The UK Government announced in May that it will invest £8.2 million to support advanced maths for girls in schools. The funding is part of a refresh to the Advanced Maths Support Programme. Designed to improve the skills base needed for the fast-developing AI sector, the programme includes pilot teacher training and student enrichment courses on the key maths concepts and skills needed for AI.

Around 7,500 girls from 400 disadvantaged secondary schools will be eligible. It is expected that 450 students and 360 teachers will benefit from this September.

The Government's focus on girls comes after research was published

by the Department for Education in March, which showed boys are outperforming girls in maths and science at school.

UCL researchers analysing the latest Trends in International Mathematics and Science Study (TIMSS) found that improvement since 2019 in maths and science was far higher in boys than girls. In testing of year 5 and year 9 students across 267 schools in England, the 2023 data showed a gap of 26 points (out of 600) between boys and girls. The last time the exercise was carried out, in 2019, the gap was just two points. The trend was common to many other countries, but England showed the greatest increase in the gap.

All-female space flight makes headlines

In April, a launch by Amazon founder Jeff Bezos's Blue Origin space programme was publicised as the first all-female space flight, when his fiancée Lauren Sanchez travelled with five friends as part of her hen party.

Critics were quick to point out that the first all-female space flight was in fact made by cosmonaut Valentina Tereshkova in 1963 (and didn't include any pop stars).

NASA's Artemis mission has set a goal to see the first woman on the moon. Its first crewed lunar landing is currently scheduled for 2027.



BBC launches 'Horrible Science' series

The team behind 'Horrible Histories' and 'Ghosts' have turned their focus to science in a new BBC children's comedy series.

'Horrible Science' launched in May and covers a range of science topics as well as several key scientific figures such as Isaac Newton, Albert Einstein and Charles Darwin.

Highest likelihood yet of life on an exoplanet

In April there were widespread reports in the media about biological markers being found in an exoplanet for the first time, leading to enthusiastic speculation about the possible existence of alien life.

Analysing data from the James Webb Space Telescope, astronomers in Cambridge detected large amounts of dimethyl sulphide (DMS) and dimethyl disulphide (DMDS) on planet K2-18b.

On Earth, these compounds are only created through biological processes.

The results are considered 99.7% accurate but still fall below the 'five sigma' threshold of certainty for the data. Several independent scientists voiced concerns that, even if the data is reliable, there may yet be other explanations for the presence of the substances on the planet.

The Cambridge team analysed K2-18b's atmosphere by measuring changes to the light from the parent star as the planet transited in front of it. See our last issue on 'Satellites and space probes' for more on how to demonstrate the process in the classroom.



Credit: Johannes Plenio

The giant electric circuit in the sky

Dr Keri Nicoll from the University of Reading discusses the Global Atmospheric Electric Circuit (GEC) and how your students can help researchers better understand it.

Electrical investigations of Earth's atmosphere date back to the experiments of Benjamin Franklin and Thomas-François Dalibard in the 1750s, when they attempted to collect electricity from lightning strikes. This led to the realisation that a 'fair-weather' electric field exists near the Earth's surface and that, even away from storms, there is a flow of charge through our atmosphere. Why this fair-weather conduction current persists remained a mystery until the 1920s, when Scottish physicist CTR Wilson suggested that the Earth has a Global Atmospheric Electric Circuit (GEC).

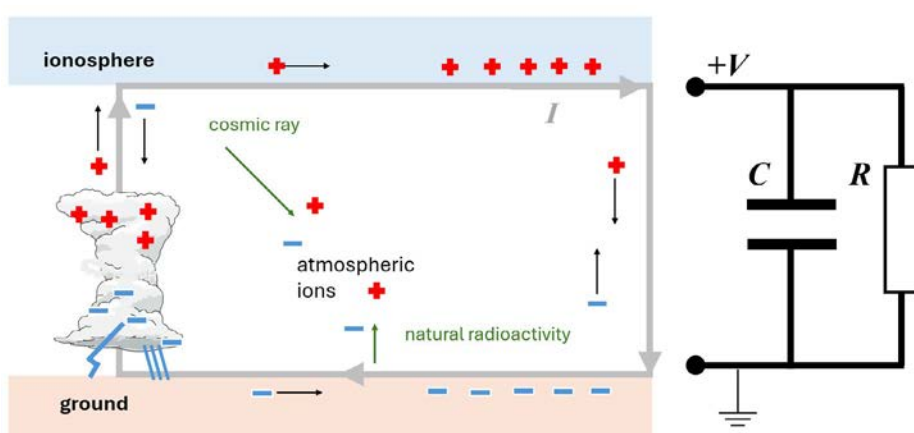
In the GEC, Earth's atmosphere can be modelled as a capacitor, with the ionosphere and Earth's surface acting as the capacitor plates. Atmospheric ions created by cosmic rays and Earth's natural radioactivity create a weakly conducting layer between the plates that slowly, but continuously, discharges the capacitor.

The system is powered by the Earth's thunderstorm clouds. The top of each cloud becomes positively charged when cloud droplets and ice particles (graupel) move in convection currents and rub against each other. This drives a current upwards towards the ionosphere. The bottom of the cloud becomes negatively charged, and this discharges as lightning or by falling as negatively charged rain to the ground. Each storm cloud is like an electrical cell with the positive terminal facing upwards. With an average of 1,000 thunderstorms at any one time on our planet, the clouds together form a giant battery

that powers the GEC by charging up its capacitor plates to an average voltage of about 250 kV.

Calling student scientists

At the Earth's surface the GEC produces an average fair-weather (away from storms) electric field of around 100 Vm^{-1} . Measuring its strength provides real-time information about atmospheric conditions, including lightning and cosmic ray activity. Looking at measurements taken over time enables scientists to analyse long-term climate trends.



Observatories in the UK have some of the longest running electric field measurements in the world, but about half a million of these historical records are still in paper form and so can't easily be analysed. This is why scientists at the University of Reading have set up a Zooniverse citizen science project – Atmospheric Electricity for Climate – to help digitise these old records. Maybe some of your students can help us unlock these valuable datasets?

Visit bit.ly/AtmosElec for more information.

Teaching tip (16–19)

To practice circuit calculations, ask students to use the values for V , C and R in the table below to show that:

- the current in the GEC is about 1 kA
- that the GEC current would drop to below 1 A in less than half an hour if all the thunderstorms on Earth suddenly stopped

(a) using V/R gives 1 kA to 2 kA
(b) assuming an initial current I_0 of 2 kA, $t = -CR\ln(I_0/I)$ gives about 1500 seconds - i.e. 25 minutes

Symbol	Quantity	Typical value (s)
V	potential difference between ionosphere and ground	200 kV to 400 kV
C	capacitance of GEC	1 F
R	resistance of fair-weather regions of Earth's atmosphere	200 Ω

Clouds and climate change

Professor Ellie Highwood, Fellow of the Royal Meteorological Society, discusses clouds – the big uncertainty in climate models.

Clouds remain one of the most challenging parts of the climate system to understand, simulate or predict. They form on the smallest of scales, yet cover vast areas of the planet at any one time, and can change in the blink of an eye. While clouds provide inspiration for artists and writers, they are a source of frustration for physicists! But understand, simulate and predict them we must, for they are not only a vital part of the water cycle but also play a significant role in determining our climate.

Anyone who has flown on a plane on a cloudy day will have experienced the greyiness below cloud on take-off, the bumpy journey through the cloud layer and the emergence into brilliant brightness above. These low-altitude clouds, made up of water droplets, reflect a lot of sunlight at visible wavelengths back to space. This is what causes the bright, brilliant view from above – you are seeing the reflected light. Of course, it also means that less sunlight passes through the clouds, so from below they look dark and threatening. These clouds increase the planet's

albedo (the fraction of sunlight reflected back to space) and cool the Earth's surface. However, there are other types of clouds, generally much harder to see and at higher altitudes, such as cirrus. These are usually made of thin layers of ice crystals. They allow sunlight to pass through and so don't change the albedo much, but instead absorb some of the infrared radiation emitted by the Earth's surface. Being colder than the surface, they re-emit only a small amount of this back to space and so act like greenhouse gases, warming the surface. Contrails, which are cirrus clouds emitted by aircraft, warm the climate by an amount approximately equal to the greenhouse gases emitted by the aircraft fuel.

The effect of any particular cloud depends on many details, including its altitude, extent or size, the size and state of the particles it is made up of, and the brightness and temperature contrast with the surface underneath. There's a very fine balance between warming and cooling. However, clouds tend to exert a cooling effect on average over the whole globe.

But, clouds change, and in particular, clouds change as part of climate change. In a generally warmer world, there is more evaporation from the oceans and the hydrological cycle speeds up. With more water vapour in the atmosphere and a warmer

surface, moist convection (hot air rising and cooling so that water vapour condenses and forms clouds) increases, and we might expect more thick, lower-altitude clouds. These would tend to cool the climate and so are a negative feedback on temperatures. However, if there was an increase in the high-altitude cirrus clouds, then there would be a positive feedback. The net result would also vary regionally, because if the clouds are over a dark surface like the ocean, they will have a bigger effect on the albedo than if over the bright desert. Cloud increases during nighttime can't affect the albedo (as there is no sunlight during the night) but can still affect the infrared radiation.

Clouds also change depending on the number of aerosol particles in the air, as these are needed to form the droplets. Cloud seeding experiments have long been used to influence cloud and rainfall formation. One current example involves attempts to seed cloud to shade the Great Barrier Reef, reducing coral bleaching events.

Given all these complexities, perhaps it's not surprising that different climate models can project different changes to cloudiness in the future. Hopefully, some of the activities in this edition's physics pull-out (pages 9–12) might inspire some of the next generation of cloud physicists.

Physics education research

In this column,
James de Winter
 (University of Cambridge and
 University of Uppsala)
 and **Richard Brock**
 (King's College London)
 highlight publications and resources
 from physics education research
 and suggest how they may be used
 to inform classroom teaching.

For this issue, Richard writes
 about one of his own papers,
 co-written with Ben Rogers and
 Liam Cini-O'Dwyer.

research@teachphysics.co.uk

Researching the impact of stories

As regular readers of Classroom Physics know, I enjoy telling stories in physics lessons. Whether it's Babson's anti-gravity institute, Newton's prediction of the end of the world, or Hooke's publication by anagram, I feel that stories enrich physics teaching.

However, along with my colleagues Ben Rogers and Liam Cini-O'Dwyer, I was curious to know if storytelling helped or hindered learning. We wanted to understand whether stories provide a hook to encourage learning, or do they distract? We were grateful to receive support from the Institute of Physics to carry out a research study, which has just been published.

Much previous research had suggested that interesting but irrelevant information in teaching material, referred to as seductive details, can impede learning. We decided to investigate the claim in the context of stories about physics, drawn for the IOP stories booklets (spark.iop.org/stories-physics).

We used a mixed methods design comprising a survey and interviews. In the first study, we used an online survey ($n = 101$). The survey randomised participants (16–18-year-old students in England) to watch video explanations of two physics topics: dark matter and radioactive tracers. The videos were of two types – one that included seductive details and one that did not – and we compared pre-, post- and delayed knowledge test scores. In the case of the dark matter stories, we talked about the life of Vera Rubin (Rubin advised: “Don't let anyone keep you down for silly reasons such as who you are. And don't worry about prizes and fame”). In the tracers story video we discussed how de Hevesy used radioactive tracers to see if his landlady was reheating leftovers.

Six participants were interviewed. In the second study, nine participants watched the videos whilst commenting on their perceptions of learning.



Credit: Shutterstock

From the first study, we found that the seductive details in one topic (dark matter) harmed learning (those who watched the video with stories recalled less than those who watched the video without stories). For the tracers video, there was no difference between the two videos. In the second study, we used an approach called a ‘think aloud’ in which we asked students to watch videos (with and without stories) and asked them to reflect on how the content impacted their learning.

The data from the think aloud task indicate an explanation for the different effects of the stories in the two topics. Previous work had suggested three ways in which seductive details can harm learning: distraction (attention being drawn to seductive details rather than the intended material); disruption (the target argument is interrupted by unnecessary details); and diversion (the seductive details prime inappropriate knowledge – for example by making you think about Newton's work on alchemy rather than his laws).

Informed by cognitive theory, we proposed three ways in which stories can support learning: by promoting attention to core content; chunking material into more manageable units; and providing structure into which new content can be anchored when prior knowledge is low. Whilst we acknowledged some ways that stories can support learning, we suggested that stories should be used carefully. Teachers should bear in mind that stories have the potential to harm learning (at least in the kind of short-term knowledge tests we carried out) but, in some contexts, such as when students have relatively low prior knowledge, the negative impact of stories on learning may be limited.

Reference

Brock, R., Cini O'Dwyer, L., & Rogers, B. (2025). The Effect of Seductive Details on Knowledge Retention in Physics Videos: A Mixed Methods Study, *Journal of Research in Science Teaching*

Condensation and convection

Clouds and wind

Inside this pull-out:

- Experiment 1: Cloud in a beaker
- Experiment 2: Dew point temperature
- Demonstration: Why the wind blows
- Activity sheet: Ping pong ball anemometer



The physics of our weather and climate

School physics introduces many of the concepts needed for a basic understanding of our atmospheres and oceans. However, many at age 16 struggle to apply their understanding to weather or climate. For example, in a recent survey of school-leavers in England, when asked why sea levels are rising, most couldn't identify the expansion of water as a major contributor (Climate Literacy Survey, Department for Education, 2024).

Those of your students that study geography may develop a better level of climate literacy, but the different teaching approaches in the two subjects can still be a barrier. In physics, for example, students learn that atmospheric pressure is the force per unit area due to the weight of the air above them, but may be left with the impression that it is an unvarying quantity. In geography, atmospheric pressure is usually linked to regions of highs and lows caused by the circulation of the atmosphere.

The activities in this pull-out are designed to help bridge the gap. Our cloud-themed experiments on page 10 can be used to add a weather context to the teaching of evaporation and condensation. Our demonstration on page 11 links convection to wind direction and air pressure, and the activity sheet on page 12 provides instructions for students to build their own anemometer. On this page are some other physics topics that can be linked to the geography curriculum through weather and climate contexts.

See bit.ly/climate-literacy for more information about climate literacy.

Professor Sylvia Knight is Head of Education at the Royal Meteorological Society.

Physics topic	In context	Geography curriculum link
Expansion of liquids	The water in our oceans expands when its temperature increases.	Sea-level rises are evidence of temperature increases caused by global warming.
Seasons	The Earth's spin axis is tilted at 23.5°, and the Earth's orbital path is nearly a circle (eccentricity of 0.016). These do not change significantly over a year or even decades, but they do vary over thousands of years because of the gravitational pull of other bodies in the solar system.	Natural climate change can happen because of Milankovitch cycles, which are: variations in the eccentricity of the Earth's orbit; precession of its spin axis; and changes in its tilt angle.
Specific heat capacity	Water has a higher specific heat capacity than plants, soil and rock. For the same rate of heating, the temperature of land rises more quickly than the same mass of ocean.	Continental climates are more extreme than maritime ones.
Specific latent heat	Evaporation of water from the sea and condensation into clouds transfers energy from the sea to the atmosphere.	Storms have the fastest wind speeds when they are over or near the ocean.
Electromagnetic waves	The land and oceans are heated by absorbing visible light and short wavelength infrared radiation from the Sun. The Earth cools by emitting longer wavelength infrared radiation which is absorbed by some gases in the atmosphere, such as carbon dioxide, methane and water vapour.	Human activities such as burning fossil fuels produce carbon dioxide and methane, which enhance the Earth's natural greenhouse effect.

Experiment 1: Cloud in a beaker

In this activity, students use water vapour and smoke particles to make a cloud in a beaker.

Equipment (per group)

- Beaker
- Aluminium foil
- Crushed ice
- Access to hot water
- Match and means of lighting it (or you may want to provide this when the students are ready)

Procedure

Introduce the activity by explaining that cloud droplets are water droplets suspended in the air that form when water vapour condenses onto small dust, pollen or soot particles.

Ask students to:

1. Make a lid for the beaker using the aluminium foil and place crushed ice onto it.
2. Pour a small amount of warm water into the beaker to increase the water vapour content of the air inside.
3. Check the lid feels very cold to the touch. Wait a few minutes if required.

4. Throw a lit match into the water (or ask the teacher to do this) and then quickly place the lid on top of the glass.
5. Observe the inside of the beaker. (They should see a cloud form near the top of the beaker – they may also see it rain.)

Explanation

Cloud droplets form because water vapour rises to the top of the beaker, cools, and condenses onto smoke particles. If the droplets grow too big, the rising air can no longer support them and they fall as rain.

Teaching tip

To help students visualise a cloud droplet and develop a better understanding of the forces involved, demonstrate a ping pong ball suspended in the air using the updraft from a hairdryer.

Cloud seeding

By adding particles to the air it is possible to seed clouds and increase levels of rainfall or snow. Find out more: bit.ly/STEMCloudControl

Experiment 2: Dew point temperature

In this activity, students measure the temperature at which cloud droplets form out of the air.

Equipment (per group)

- Empty, clean tin can (e.g. baked bean tin) with the lid removed, leaving no sharp edges
- Thermometer
- Crushed ice
- Salt (optional)

Technical notes

If the air in the room is dry, the dew point can be below the freezing point of water. If no water droplets form when the temperature reads 0°C, students can add salt and ice to lower the temperature.

Procedure

Introduce the activity by explaining that the temperature at which water vapour in the air condenses to form water droplets on a solid surface is called the dew point.

Ask students to:

1. Put the thermometer in the can and half fill it with water.
2. Add a little ice and wait for the temperature to fall. Stir and wait for the ice to melt.

3. Repeat the previous step until condensation is seen forming on the outside of the can.
4. Note the reading on the thermometer. This is the dew point temperature.

Explanation

Individual water molecules are continuously evaporating and condensing at the surfaces of the can. At room temperature, the evaporation rate is higher than the condensation rate, but lowering the temperature reduces the evaporation rate. At the dew point, the evaporation rate becomes lower than the condensation rate, and water molecules start collecting on the surface of the can.

More resources from the Royal Meteorological Society

For other experiments, demonstrations and DIY weather instruments to bring weather and climate teaching to life, visit metlink.org/experiments

For a collection of self-marking physics questions for students on weather and climate, produced in collaboration with Isaac Physics, visit metlink.org/resource/isaac-physics

Demonstration: Why the wind blows

In this demonstration, students see how surface winds such as sea breezes form.

Procedure

Introduce the demonstration by explaining that when sea and land are heated by the Sun, the temperature of the land increases more quickly than the sea. If students are familiar with the idea of specific heat capacity, you could compare values for sand ($830 \text{ J kg}^{-1} \text{ K}^{-1}$) and water ($4200 \text{ J kg}^{-1} \text{ K}^{-1}$).

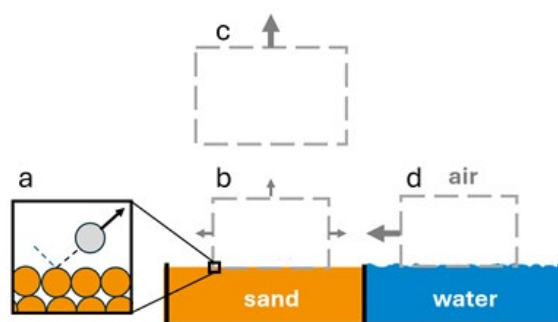
1. Pour ice-water into a glass dish. This represents the cold sea.
2. Switch on a hot plate and set the temperature to 80°C .
3. Place sand into a metal bowl and on to the hot plate. Stir with a wooden spoon for even heating.
4. Pour the hot sand into another glass dish. This represents the hot land.
5. Place the two dishes next to each other and make a cardboard enclosure to stop any breezes in the room interfering with results.
6. Light a paper straw, blow it out, and hold between the dishes near their surfaces. The direction in which the smoke moves shows the direction the air is moving.
7. Ask students why they think a wind blows from sea to land.

Teaching notes

A common misconception is that heat is a substance that moves through the air. Support students in developing an explanation of why the air moves from sea to land based on temperature, volume, density and particle speeds. For example:

- a. air molecules speed up when they collide with fast-vibrating atoms in the sand
- b. air temperature increases, making air expand and so reducing its density
- c. less dense air rises
- d. cooler air moves towards the low density region to replace the rising air

Surface winds are the ground-level manifestations of atmospheric convection cells that also drive cloud formation.



Weather systems: Highs and Lows

Weather forecasters and climate scientists often talk about Low- or High-pressure systems bringing unsettled or settled weather. This is because wet weather systems form when water vapour rises, cools and condenses to form clouds. At ground level, friction slows the air that moves in to replace the rising air, which reduces the mass in the air column above that location, and so surface pressure falls. Conversely, systems of descending air that normally bring settled weather are known as High-pressure systems because friction at ground level slows the air that is moving out of their bases and so surface pressure rises.

Weather system	Motion of air	Likely weather conditions
Low-pressure	Upwards	Cloudy/rain
High-pressure	Downwards	Clear skies

Clouds calculations

The temperature in the troposphere reduces linearly with altitude by about 6.5°C per km and so the height h at which the base of the lowest clouds can form on a particular day can be estimated using:

$$h \text{ (in km)} = \frac{\theta_s - \theta_d}{6.5^\circ \text{C}}$$

Where θ_s is the outdoor temperature at the Earth's surface, and θ_d is the dew point of the air, in degrees Celsius. Values for both for today are available on weather websites or apps.

Physics topic	In context	Geography curriculum link
Convection	Our planet has giant convection cells that span thousands of kilometres across the surface and to over 10 km up into the atmosphere. At the surface these create belts of Low and High atmospheric pressure that wrap around the planet.	Earth's desert climates can be found at the High-pressure belts and its wet climates at the Low-pressure belts. Global winds along our planet's surface, such as the trade winds, blow between belts.

Ping pong ball anemometer

Wind speed is how fast air moves over the ground. It can be measured using an instrument called an anemometer and described using the Beaufort scale.

You will need:

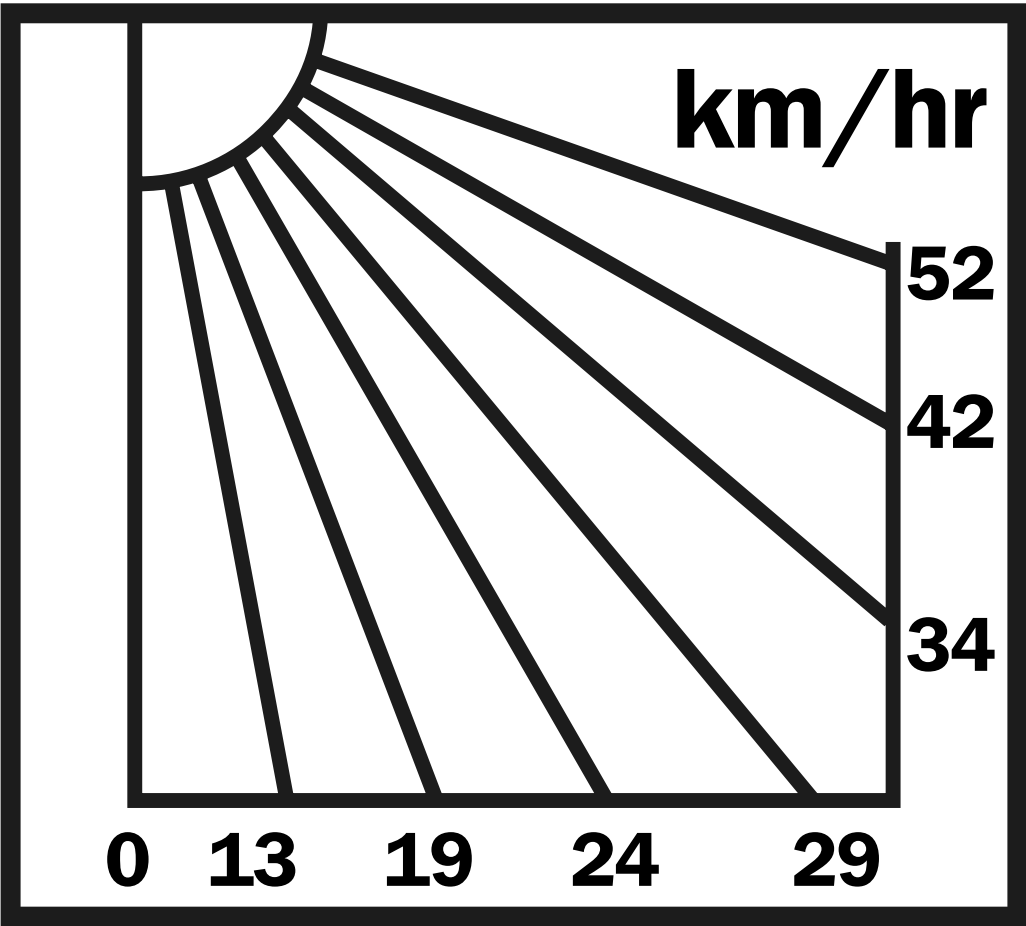
- Strong thread
- 30 cm ruler
- Ping pong ball
- Scissors and sticky tape
- A ruler
- Clamp and stand
- Access to a hairdryer or fan

To make your ping pong ball anemometer:

1. Cut out the wind dial (below).
2. Measure and cut 30 cm of thread.
3. Use sticky tape to attach one end of the thread to a ping pong ball and the other end to the back of your dial.
4. Stick the top of your dial to the ruler.
5. Clamp the ruler in position so that the string is vertical and points to zero.
6. Test your anemometer by blowing gently on it.
7. Measure the wind speed of a hairdryer/fan.
8. Use the table above to identify the wind speed descriptions for your measurements.

Wind Speed	Beaufort Number	Description
0-1 km/hr	0	Calm
2-5 km/hr	1	Light air
6-11 km/hr	2	Light breeze
12-19 km/hr	3	Gentle breeze
20-29 km/hr	4	Moderate breeze
30-39 km/hr	5	Fresh breeze
40-50 km/hr	6	Strong breeze
51-61 km/hr	7	Near gale
62-74 km/hr	8	Gale
75-87 km/hr	9	Strong gale
88-101 km/hr	10	Storm
102-117 km/hr	11	Violent storm
>117 km/hr	12	Hurricane

Wind dial



Atmospheric stories

Meteor mirroring

Many readers will be familiar with the phenomenon of shortwave radio waves reflecting from the ionosphere, a layer of charged particles in the atmosphere. Before the introduction of satellite technology, ionosphere reflection 'skywave' communication was used to send radio waves over long distances. Other areas of ionisation can also cause reflection. During its motion around the sun, the Earth's atmosphere strikes small rocky particles, which we call meteors. They burn up, and the heat of their combustion creates trails of ionisation in the atmosphere which can reflect shortwave radio waves. The effect, first noted in Japan in 1929, typically lasts a few seconds. Several military projects (JANET in Canada in the early 1950s, and NATO's COMET in the 1960s) sought to make use of meteor-scatter for communication. For over 40 years, snow water content gauging stations in the US used signals that relied on meteor-scatter. Whilst the approach is cheaper than satellite communication, the lower bandwidth means it has largely fallen out of use.

How high is the sky?

The edge of space is demarcated by a boundary called the Kármán line (or sometimes the von Kármán line). The boundary of space was first proposed by Theodore von Kármán who calculated a theoretical maximum altitude for aircraft flight of 83.8 km above the Earth's surface. In general, it is assumed that there is no clear division between the atmosphere and space; the characteristics of the atmosphere change gradually and there is no distinct boundary. The international aviation body, the FAI (Fédération Aéronautique Internationale), had set the altitude of the Kármán line at 100 km but, in 2018, proposed reducing it to 80 km. This definition coheres with the awarding of astronaut status: in the US, to qualify for astronauts' wings, military personnel must reach an altitude of 80 km. Civilian standards for becoming an astronaut are more stringent. Under Federal Aviation Administration (FAA) guidance a civilian must reach an altitude of 80 km and must also have "demonstrated activities during flight that were essential to public

safety, or contributed to human space flight safety".

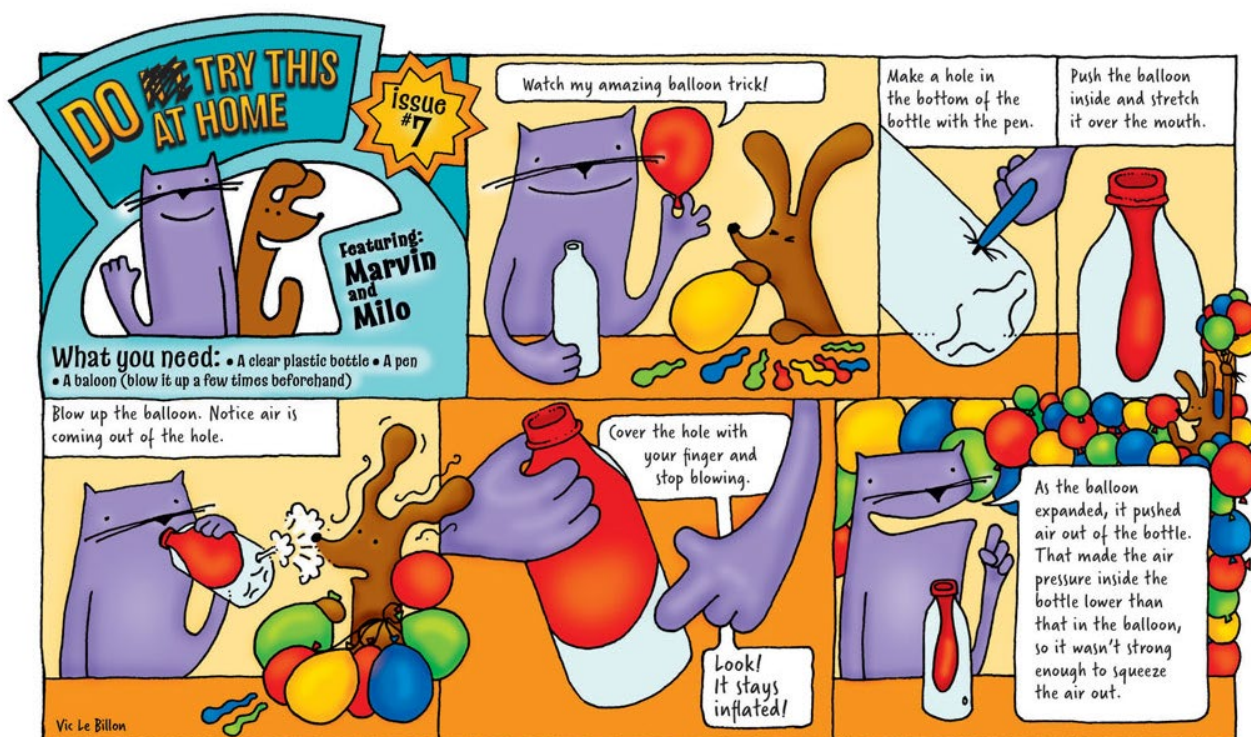
Halley's alternative atmospheres

The second Astronomer Royal, Edmund Halley, is best remembered for his eponymous comet, however he had curious views on the nature of the atmosphere. In 1692, Halley proposed that the Earth was formed of a hollow shell 500 miles in diameter. Within the shell were two further concentric shells, and in the centre a core. A layer of atmosphere separated each shell. He proposed that the Aurora Borealis was caused by gas from inner atmospheric layers escaping. In a painting of Halley at the age of 80, he is depicted holding an image of the layered Earth model, leading one historian to infer that he felt the hollow Earth was his most significant contribution.

spark.iop.org/stories-physics

Compiled by Richard Brock.

Follow him on X (Twitter) at [@RBrockPhysics](https://twitter.com/RBrockPhysics)



Download more Marvin and Milo activities at iop.org/marvinandmilo

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Physicseducation

Physics Education is our international online journal for everyone involved with the teaching of physics in schools and colleges.

Editor-in-chief Gary Williams highlights his favourite papers on **atmospheric physics** from the archive and shares some highlights from the current volume.

Access over 50 years of articles at iopscience.org/physed

Affiliated schools have free access – email affiliation@iop.org for a reminder of your log-in details.

Having purchased a few Goethe's barometers as presents and been suitably impressed with them (as have the recipients), I can easily recommend the paper "An equation for the 'weather glass'" by Michael Jewess. Often made of glass in decorative forms, a trapped bubble of air pushes liquid up a thin neck when the outside air pressure is less than the inside air pressure, and vice versa. Sometimes the difference is enough for the liquid to drip out. This paper gives some history to atmospheric pressure measurement and describes how to go further than the largely qualitative decorative type design. These barometers can be purchased as an interesting item for a desk, but can also be made easily with a bottle, some flexible pipe and a length of glass tube.

Quite a few schools have now sent balloons up to the edge of space to get camera shots of the curvature of the earth and see the edge of the atmosphere. The open access paper "Experiments at the edge of space: balloon flights to the stratosphere" takes this further, describing how to take temperature, particle and radiation data from the stratosphere using relatively inexpensive off-the-shelf apparatus. The story of weather is both fascinating and relevant to us all and this quite major project brings opportunities for exciting physics with an added feeling of adventure.

The small computers for sense and control that are Arduinos have a multitude of uses. From inexpensive Hall probes to pH sensors and light gates, they seem to have no end of applications. In this pair of papers, "Building a digital barometer with Arduino to study forces due to atmospheric pressure and weather maps at schools" and "Development of a micro weather station using Arduino and Internet of Things", you can read about a very simple digital barometer suitable for students aged 11–14 and a far more advanced weather station with temperature, humidity and pressure sensors which connect online. Making a temperature and humidity monitor is relatively easy, even if you don't build a complete station. See later in the article for more help with Arduinos.



A Goethe's barometer

"An equation for the 'weather glass'" bit.ly/PEWeatherGlass

"Experiments at the edge of space..." bit.ly/3yo3pTn

"Building a digital barometer..." bit.ly/PEDigitalBarometer

"Development of a micro weather station..." bit.ly/PEMicroWeatherStation

Recent articles

Inexpensive investigations like the one described in the paper "How cardboard gravity-powered ramp walker really works" are always valuable. This simple toy, which is made from card and 'walks' down a ramp needing only a timer to get data, looks like fun too. The paper describes the mechanism in detail but this idea would be suitable from primary up to tertiary levels by changing the depth of the investigation and explanation.

Many readers will have seen the demonstration using a tall glass container, like a gas jar, inverted and placed over a candle, sitting in water such that the pressure of the atmosphere can push the water up inside the container. In the paper "Joint measurement of temperature, pressure and water height as a function of time in the candle covered by a glass experiment", there are some interesting ideas on how this can be taken further, so you can start with a demonstration and make it into an investigation.

It is highly likely that you will come across language issues when teaching. "Silent physics animation

tasks for learner narration” presents a useful solution to language resource issues by using silent movies that students add their own narration to. You can show one movie, and students add narration in the language of their choice. The suggestion is that after the narration, the students and teacher can engage in a discussion, going through explanations and introducing technical language that students need to become familiar with. This seems an excellent idea and avoids a need for translations.

Credit: IOP



The candle covered by a glass experiment

“How cardboard gravity-powered ramp walker really works”
bit.ly/PECandleGlass

“Joint measurement of temperature, pressure and water height...”
bit.ly/PECandleGlass

“Silent physics animation tasks for learner narration”
bit.ly/PESilentPhysics



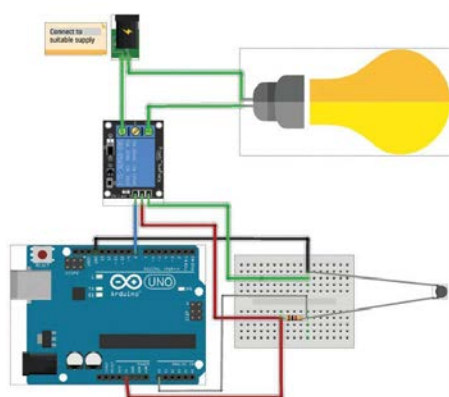
Cardboard ‘walkers’

Credit: IOP

From the archive

Returning to the atmosphere theme, I recall Sylvia Knight from the Royal Meteorological Society casually dropping into a telephone call the fact that “clouds are in layers”, which seemed obvious when I thought about it, but up until she pointed it out I had never given it much thought! Sylvia shows this, in her paper “Experiments demonstrate clouds forming in the classroom”, by making clouds in a bottle. She also co-authors a paper, simply called “Clouds”, which describes in much more detail how clouds form. The paper would be suitable for a class handout for 14–18 year-olds and covers the temperature of different layers of the atmosphere, different cloud types, unusual clouds, and weather systems.

Credit: IOP



Arduino use for beginners

Returning to another previously mentioned theme, the Arduino: should you wish to take advantage of the flexibility of these inexpensive devices but are not confident about microelectronics, then take a look at John Kinchin’s Frontline paper “Using an Arduino in physics teaching for beginners”. This paper will give you enough information to be able to construct the previously mentioned digital barometer and in the quick links you’ll find more papers from John that are aimed at those new to Arduinos. Once you get started then searching the Physics Education website will bring you hundreds of Arduino papers, from the very simple to the very advanced.

“Experiments demonstrate clouds forming in the classroom”
bit.ly/PECloudsClassroom

“Clouds”
bit.ly/PEClouds

“Using an Arduino in physics teaching for beginners”
bit.ly/PEArduinoBeginners

Physics education

Quick Links

“Using an Arduino in physics teaching: LDR as a simple light sensor” A well-explained and easy-to-do Arduino activity from John Kinchin

bit.ly/PEArduinoLDRSensor

“Using an Arduino and cheap thermistor to make a simple temperature sensor” More from John Kinchin to take things further with your Arduino skills

bit.ly/PEArduinoThermistor

“Building a temperature controlled model house using an Arduino microcontroller” An application for the circuits you already built with the Arduino!

bit.ly/PEArduinoMicrocontroller

Open access

“Exploring rotations by a modified fidget spinner and a smartphone” A good reason to buy some fidget spinners

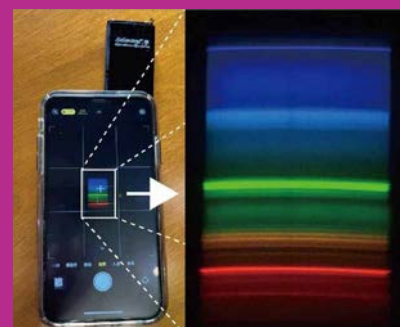
bit.ly/PEFidgetSpinnerRotations

“Arduino-based infrared power meter for use with a Leslie cube” Take your Arduino skills to the next level

bit.ly/PELeslieCube

“Teaching fluorescence of plants & algae in physics class to address climate change” Lots of interesting physics with a biology and climate change flavour

bit.ly/PEFluorescencePlantsAlgae



Fluorescence of plants and algae

Credit: IOP

physicsworld

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Astronomers create a 'weather map' for a gas giant exoplanet

In this article from February this year, Physics World contributing editor Isabelle Dumé writes about work by astronomers monitoring the exoplanet WASP-127b, discovered in 2016.

The planet's atmosphere was examined by looking at starlight filtered through its upper atmosphere during a transit of the planet in front of its host star.

Data from the 6.5-hour transit indicates that the winds on the planet reach speeds of up to 33,000 km/h. This is faster than the speed of the planet's rotation – a phenomenon not seen in our solar system. These wind speeds, the fastest seen anywhere by astronomers, made it possible for the research team to model the planet's weather system.

Lead author Lisa Nortmann from the University of Gottingen, explained: "The unexpectedly fast velocities measured in this planet's atmosphere have allowed us to investigate different regions on the planet, as it causes their signals to shift to different parts

of the light spectrum. This meant we could reconstruct a rough weather map of the planet, even though we cannot resolve these different regions optically."

bit.ly/PWExoplanetWeather

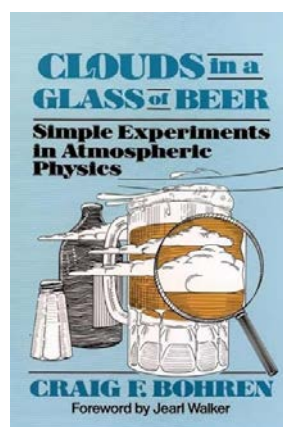
See our last issue for classroom activities based on solar transits.

Cloudy with a chance of warming: how physicists are studying the dynamical impact of clouds on climate change

This article from November 2024 takes an in-depth look at the latest thinking on how clouds are impacting climate change. (See *feature article*, page 7.) As science writer Michael Allen explains, clouds are a key driver of the Earth's 'radiation budget', or "the balance between the amount of energy coming in from solar radiation, and the amount of energy going back out to space".

bit.ly/PWClouds

For this issue, Sylvia Knight, head of education at RMetS, recommended 'Clouds in a glass of beer' by Craig F. Bohren, her favourite textbook on atmospheric physics. Ruth Wiltsher, an IOP teaching coach, had a read.



Book corner

'Clouds in a glass of beer' was recommended by Sylvia Knight, who said that, despite its age, it's the best book on atmospheric physics for teachers or sixth formers. She told us: "I don't generally like popular science writing because it usually leaves me feeling that I haven't been given enough information to really understand what is being described; this book is an exception to that rule."

Published in the US in 1987, with reprints in 2001 and 2003, the book is showing its age, with black and white drawings and photos throughout – but as Sylvia points out, the atmospheric physics remains sound. It's likely you will need to borrow a copy or buy second hand (ex-library copies are available online).

Divided into humorously titled chapters, including 'A Murder in Ceylon', 'Physics on a Manure Heap'

and 'Multiple Scattering at the Breakfast Table', the book contains 25 simple and interesting experiments that reproduce natural phenomena with inexpensive materials. These sit alongside explanations of the physics concepts behind them.

As well as discussing clouds, Bohren takes the opportunity to debunk some prevalent physics myths, like why, on a clear day, you can't see forever. He also explains the differences between concepts such as sun light, sky light and air light.

Bohren's approach is to use words, and the odd simple graph, diagram or photo, but without complex equations that need a lot of explaining. Through anecdote, he provides examples of the processes being discussed which make them easier to visualise as well as more relatable and, ultimately, memorable.

bit.ly/CloudsGlassofBeer



Create an effective learning atmosphere for younger secondary school students with these materials on Earth's atmosphere and air pollution from RSC Education.

edu.rsc.org



A down-to-earth approach to teaching

At the Royal Society of Chemistry, we have developed resources to help teach your 11–14 year-old learners about the composition of Earth's atmosphere. While we can't guarantee 'stratospheric' success, we can guarantee useful posters and classroom activities that get to the burning issues of climate change, the greenhouse effect, acid rain and smog.

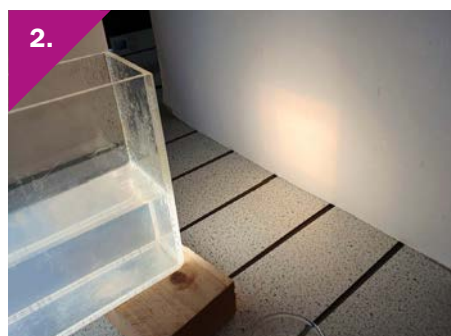
Suitable for key stage 3 and S1–3 classes, these downloadable pdf posters, supporting fact sheets and classroom-ready activities are authored and quality controlled by classroom teachers and educators. They provide relatable context for learners, in the form of Did-you-knows and illustrations, making some of the more difficult-to-grasp details relatable.

The Earth's atmosphere collection (rsc.li/42xP2rU) includes a poster

showing the layers, composition and key facts about the atmosphere, alongside a fact sheet providing learners with more details. Learners use these to complete a DART (directed activity related to text) worksheet – you can pick from three levels of scaffolding – including a reading comprehension, creative writing and data-handling activities.

With the Earth's atmospheric pollution collection (rsc.li/3FWAidj), you get a poster covering the changes human activity has brought about, that details which chemical compounds are involved. The fact sheet provides learners with more information on climate change, acid rain and smog. You can use the classroom-ready activities to support learning about pollution, alongside developing pupils' scientific literacy skills and helping them critically evaluate information sources.

Nick Mitchener RSciTech, physics technician at Ferndown Upper School in Dorset, and a recipient of the 2024 IOP Technician Award, gives a simple demonstration for Rayleigh scattering.



Why is the sky blue?

We see the sky as blue and the sun as yellow/orange, and then at sunset the sky goes orange and the sun goes a darker orange and sometimes red. Occasionally, we get dust clouds of fine sand blown up from the Sahara and the sky goes orange.

All of this is due to Rayleigh scattering, the process by which light from the sun is scattered by gases and particles in the atmosphere. The shorter blue wavelength is scattered and we see that as the sky. However, when we look at the sun, we see it as yellow, orange or red. The blue has been scattered away by the higher atmosphere, so we see the longer yellow, orange and red wavelengths, which are scattered away less than the blue.

We can demonstrate this with a bright light and a little bit of milk in water. In the pictures I have used a trough and a slide projector. If you don't have a trough you can use a large beaker.

A ray box will work as a light source. If you use a raybox, it is unlikely to project, but you can get students to view the effect through the milky water.

It takes very little milk to get the blue effect, then as you add more milk you will start to see the end of the tank going orange. Use a bit of trial and error to find out what works for you.

It is a very simple and classic demo, but it helps answer a common question.

Photos (left):

1. The setup: an acrylic tank of water with a screen at the end, a slide projector with a halogen bulb, and a beaker of milk with a pipette.
2. After the milk is added, while the tank looks a bit blue, the light projected onto the screen starts to change to yellow as the blue is scattered away.

For more information on IOP Awards for teachers and technicians, visit iop.org/about/awards/



Samir Moezzi from CLEAPSS revisits a STEM investigation from 2016 involving a high-altitude weather balloon.

www.cleapss.org.uk



Clarissa the CLEAPSS bunny reaches high altitude

Watch footage from the investigation at youtube.com/user/CLEAPSS

A STEM project reaches new heights

Back in 2016, CLEAPSS investigated a popular STEM project involving a high-altitude weather balloon. During the flight, we captured photos and video footage, tracked the payload, and successfully retrieved it afterwards.

Our payload included an action camera and external power bank, a tracking system and various electronics and sensors. To improve coverage, we programmed a Raspberry Pi to take pictures at regular intervals and transmit them using a homemade aerial. The Pi was also connected to a GPS module, allowing it to send live telemetry data including altitude, speed, temperature and location.

As backup, we included both a basic GSM GPS tracker (SIM card-based) and a SPOT GPS tracker, with operating ceilings of a few hundred metres and up to 18km, respectively. The Raspberry Pi continued functioning throughout the entire flight. We also included our mascot, Clarissa the CLEAPSS bunny!

This investigation provides ample opportunities to study atmospheric physics. An addition we would consider in the future would be radiation detection, to examine variation in count rates at different altitudes.

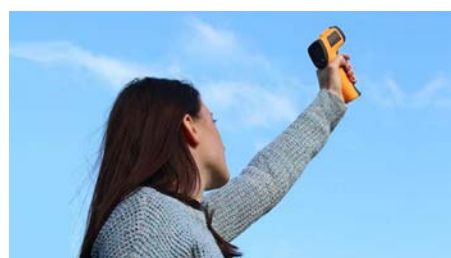
We acquired a balloon kit which included essentials such as the balloon itself, a polystyrene payload box, parachute, radar reflector, tethering string, and more. We sourced balloon gas (helium) from BOC Gases.

At least 28 days before launch, you must notify the Civil Aviation Authority (CAA) via an online form. It's a good idea to submit a range of potential launch dates in case of changes in wind conditions. Weather prediction tools like predict.habhub.org can estimate your balloon's trajectory using current weather models. Forecasts are more reliable during the final 72 hours before launch.

On launch day, set aside up to 3 hours for setup. You may be required to contact local airfields for up-to-date airspace information. Be sure to bring zip-ties, duct tape, scissors, Newton meters (for measuring lift force), and plenty of patience! The balloon's ascent rate – determined by the upward pulling force – is critical to predicting the landing site.

Upon release, you'll likely lose sight of the balloon within a few minutes. From that point on, tracking data becomes crucial. Back at CLEAPSS HQ, we followed the live telemetry with great excitement – like our own version of ground control!

IOP Fellow, former physics teacher coach and T³ Europe STEM lead Ian Galloway writes about an easy experiment to measure the temperature of different cloud types.



Reach for the sky

Infrared thermometers are inexpensive (from about £10 online) and, courtesy of the Covid pandemic, they are now ubiquitous. Using them to study the temperatures of different cloud types is a great way to spark curiosity and stimulate interest in the weather.

All that is required is to point the thermometer at a cloud on a sunny day with clearly defined cloud types.

It may come as a surprise to most students to find that the clear blue sky is so cold – well below the range of most IR thermometers – whereas cloud temperatures are

easily measurable and so could form the basis of an investigation. Data published by NASA (go.nasa.gov/3Sajoeh) could be used as the basis for a worksheet and source of information.

Other useful references are on YouTube (bit.ly/YTInfrared and bit.ly/YTInfrared2) and the BBC (bbc.in/3YPNqrn).

IOP Spark has a similar activity relating to measuring greenhouse gases at spark.iop.org/



Science and
Technology
Facilities Council

A particle physics course for high school students

CERN's physics education research team recently published the pilot version of a particle physics course for high school students.

With 16 chapters containing videos and quizzes, the course covers a core curriculum that introduces students to fundamental questions, such as "What is a particle?" or "What are charges and interactions?". These lead to more applied questions, such as "What is a particle accelerator?" and "What is a particle detector?".

The total length of the course videos is close to four hours, and the pilot version features instructions for DIY experiments that can be performed at home. The course is free and open to everyone.

Once you complete all chapters by answering the corresponding quiz questions correctly, you receive a digital certificate!

Curious to try it out yourself?
Go to **cern.ch/PPC** and start the course now.

Develop physics subject knowledge, pedagogy and confidence in the classroom

FREE CPD from The Ogden Trust for teachers in the English state sector teaching physics at KS3/4 without a physics specialism

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SCIENCE MUSEUM GROUP

What happens to you in space?

Have you ever wondered what would happen to an astronaut's body in space if they weren't wearing a spacesuit?



This activity from the Science Museum Group uses a marshmallow to show what happens when normal atmospheric pressure is reduced. The marshmallow expands dramatically... but what's the science?

Download the activity to create your own vacuum in a bottle and investigate the awesome power of atmospheric pressure:

learning.sciencemuseumgroup.org.uk/resources/no-pressure/



MAKING 	Age 11-14	Topic EARTH AND SPACE	 20 MIN
	Skills used MAKING OBSERVATIONS • CURIOSITY		



project earth

Project Earth empowers young people to innovate, offering support and advice to bring their ideas to fruition, to amplify ideas and to showcase phenomenal creativity and imagination.

Accelerator initiatives like this generally focus on young people between 18–35. We focus on young people between 8–18 because we know they have huge potential and want agency in shaping their future. We have also found in the pilot, which involves technological innovation, social transformation and cultural exchange, that tackling problems collaboratively helps to alleviate climate anxiety.

If you submit your idea then we'll start working on supporting you. There are opportunities for you to pitch your ideas at an exciting demo event in June 2026.

We'd love to work with you. Get involved via the website, **projectearth.global** or email **becky@projectearth.global**

Project Earth is affiliated to the Earthshot Prize and supported by a phenomenal group of expert advisors **projectearth.global/advisors/**

Upcoming events...

Full details via spark.iop.org/events

IOP Teaching Physics (16–19) CPD events

Nuclear Physics

3 June, 5.30pm–6.30pm

Astrophysics

10 June, 5.30pm–6.30pm

Visit bit.ly/Physics1619 for more information

Birmingham Physics Community Day Conference

25 June, all day

University of Birmingham

Visit bit.ly/BhamPhysicsDay25 to register

IOP Wales/Royal Society of Chemistry Wales Summer Conferences

Tuesday 1 July

Pontypridd

Friday 4 July

Bangor

Welsh Physics Teachers Conference

Friday 3 October

Brecon

Royal Society Summer Exhibition Quantum Zone

1–6 July

The Royal Society, London

Part of the UN international Year of Quantum, coordinated by the IOP

Find out more: bit.ly/QuantumZone

Mimi's Tiny Adventure

1 July–31 December

Around the UK and Ireland

Details: bit.ly/MimisTinyAdventure



Physics: Basics to Brilliance programme

12 June–4 July

ASE and Physics Partners' dynamic, online, interactive workshop series is designed to provide subject-specific support for GCSE Physics teachers.

Select from six 90-minute modules covering:

Force Unveiled, Energy Explored, Electrifying Circuits, Wave Wonders, Magnetic Mysteries and The Heat is On.

Discounted for ASE members.

Find out more at ase.org.uk/basics-brilliance

See all ASE events at ase.org.uk/events



'Festival of Physics' CPD days

Friday 20 June

University of Leicester

Saturday 5 July

Winchester College

Discover hands-on workshops, expert-led sessions, and inspiring keynotes at these unmissable days of CPD for all secondary physics teachers, trainees and technicians. Tickets cost £10 and include lunch, resources, a free raffle and a packed programme of engaging activities.

Visit physicspartners.com/ for more information and to book, or email info@physicspartners.com

Seen elsewhere...

History of the barometer

An entertaining animation from TEDEd about the history of the barometer, and how it works.

bit.ly/YTBarometer



Credit: TEDEd

Why no two people see the same rainbow

A detailed look at refraction and how rainbows form, from YouTuber Veritasium.

bit.ly/YTRainbows



Credit: Veritasium

Technical Education Resource Hub

Everything you need to know about technical education pathways, in a new resource from the Careers Development Institute.

bit.ly/CDITechnicalEducation



Neon Collections

Inspire your students with Neon Futures! Neon's new collections give educators step-by-step support to engage students with engineering and technology careers.

Check out the collections now: neonfutures.org.uk/collections/